

FairLens AI Bias Audit Report

Audit ID: FL-2026-046068

Date: May 03, 2026

Submitted by: CSV Upload User

Executive Summary

This audit assessed an AI-powered loan approval system for bias, discrimination, and regulatory compliance. The system received an overall risk rating of **HIGH RISK** ■ with a composite fairness score of **12.0/100**.

The model exhibits significant bias and discrimination risks. Immediate remediation is required before any production deployment.

Regulatory Context

This audit evaluates compliance with:

- **Equal Credit Opportunity Act (ECOA):** Prohibits discrimination in credit decisions
- **EU AI Act Article 9:** Classifies credit scoring as high-risk AI requiring strict oversight
- **Fair Credit Reporting Act (FCRA):** Regulates use of consumer credit information
- **GDPR Article 5:** Requires data minimization and privacy protection
- **Federal Reserve SR 11-7:** Model risk management guidance

1. Discrimination Detection Results

■ *Disparate Impact Analysis*

Disparate Impact Ratio: 0.12

Statistical Parity Difference: 0

Equal Opportunity Difference: 0

Interpretation:

Analyzed 3 protected attributes. Minimum DI ratio: 0.120

■ *Gender DI Analysis*

Disparate Impact Ratio: 1.0

Group Approval Rates:

- Male: 60.0% (reference/highest)
- Female: 60.0% (lowest)

DI = 1.0 ■

■ *Ethnicity DI Analysis*

Disparate Impact Ratio: 0.12

Group Approval Rates:

- White: 100.0% (reference/highest)
- Asian: 100.0%
- Hispanic: 62.5%
- Other: 25.0%
- Black: 12.5% (lowest)

DI = 0.12 ■

■ *Zip_Code DI Analysis*

Disparate Impact Ratio: 0.43

Group Approval Rates:

- 10001: 87.5% (reference/highest)
- 10002: 75.0%
- 10003: 50.0%
- 10004: 50.0%
- 10005: 37.5% (lowest)

DI = 0.43 ■

2. Accountability & Governance

Governance Maturity Level: Defined (Level 2)

Maturity Score: 60/100

Risk Level: ■ MEDIUM

RACI Matrix

3. Data Privacy Compliance

4. Remediation Recommendations

Audit Trail

Audit ID: FL-2026-046068

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Model Version: 1.0.0

Next Review Date: 2026-08-03

Composite Risk Score: 12.0/100

Overall Risk Level: HIGH RISK ■

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